

Conservation of energy, power and efficiency

Energy
Paper 1 · Combined and Separate
21 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know.

	QUESTION	ANSWER	✓
1	Define power.	The rate at which energy is transferred (or work is done).	☐
2	State the standard unit of power.	Watt (W)	☐
3	State the standard unit of time.	Second (s)	☐
4	State a unit that is equivalent to 1 W.	1 J/s	☐
5	State the term for energy transferred to stores that are not useful.	Wasted energy	☐
6	State how unwanted energy transfers can be reduced.	By lubrication and by thermal insulation.	☐
7	Lubricating moving parts reduces unwanted energy transfers to which energy store?	The thermal energy store	☐
8	Describe what it means for a device to be efficient.	Most of the input energy is transferred usefully; little is wasted.	☐
9	State the unit of efficiency.	It has no unit (it is a ratio).	☐
10	State how the efficiency of a mechanical energy transfer can be increased.	By lubricating the moving parts.	☐
11	State how the efficiency of a heating energy transfer can be increased.	By adding thermal insulation.	☐
12	Torches A and B each have an input power of 50 W. A has a useful output of 40 W and B of 45 W. Which is more efficient, and why?	Torch B. It transfers a greater fraction of the input power usefully (90% against 80%).	☐

QUESTION		ANSWER	✓
13	Heaters A and B each have a useful output of 2000 W. A has an input power of 2500 W and B of 2600 W. Which is more efficient, and why?	Heater A. It wastes a smaller fraction of the input power (80% efficient against 77%).	☐
14	State the equation linking power, energy transferred and time.	power = energy transferred ÷ time ($P = E \div t$)	☐
15	State the equation linking power, work done and time.	power = work done ÷ time ($P = W \div t$)	☐
16	In words, state the equation for efficiency in terms of energy.	efficiency = useful output energy transfer ÷ total input energy transfer	☐
17	In words, state the equation for efficiency in terms of power.	efficiency = useful power output ÷ total power input	☐
18	State what is meant by dissipated energy.	Energy transferred to the surroundings and stored in less useful ways.	☐
19	State what eventually happens to wasted energy.	It is transferred to the surroundings, which warm up slightly.	☐
20	Two motors lift the same load through the same height. Motor A is quicker. Which is more powerful, and why?	Motor A. It transfers the same energy in less time.	☐
21	Explain why efficiency can never be greater than 1 (100%).	Energy cannot be created, so useful output cannot exceed total input.	☐